

Long-term Measurements: Towards a Longitudinal Understanding of Human-AI Interactions

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Abstract

Language models have taken on the role of a very new type of technology, by virtue of their "human-ness" and rapid integration into users' daily lives. This combination of features can introduce longitudinal risks—cognitive, developmental and socio-affective changes in humans—that might not surface during a short-term interaction, but can have lasting long-term effects on users. This forms the basis of a critical new mission for NLP: to pivot from static, short-term evaluations of text generations to long-term measurements of behavioral changes, towards a diachronic understanding of human-model interactions. In this work, we draw from measurements used in social science fields that are crucial to understand emergent phenomena in longitudinal data. We discuss how computational methods in the field of NLP need to be combined with such measurements, not only to understand long-term safety risks of human-model interactions, but to help steer model development towards positive rather than negative outcomes for users. This ability to model human behavioral shifts as a function of model interactions can facilitate online rather than post-hoc detection of problematic behaviors, and should be leveraged in alignment frameworks to mitigate long-term risks in users.

1 Introduction

For decades, the mission of NLP has been to build theories and models of human language in order to achieve human-level language understanding and generation. At this point in time, with models having largely reached this milestone,¹ the field of NLP faces a new, and more urgent mission: measuring the developmental impact of models on human users through continued interactions.

¹We acknowledge the large body of work that finds systemic failures in reasoning, logic and generation of models. This statement only refers to the sufficient level of fluency that has led to their integration into daily human infrastructure, thus leading to the longitudinal risks we put forward.

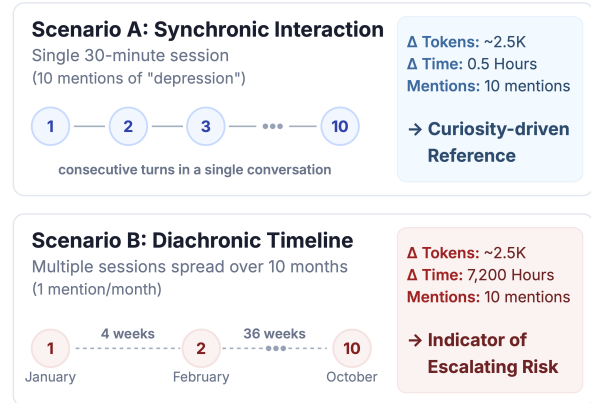


Figure 1: Two scenarios where a user's reference to depressive thoughts might indicate long-term safety risks: a single synchronic interaction (Scenario A) vs. a diachronic trend (Scenario B). The longitudinal trend in B provides a more complete understanding of high-risk user behaviors compared to single-session² evaluations.

A large part of this urgency is in reaction to how quickly these models have been adopted—and sometimes unknowingly integrated—into users' everyday lives (Susser, 2019). This paradigm shift of seamless integration of AI into user applications brings about several *longitudinal* risks i.e., ones that arise over the course of sustained interactions with systems. When coupled with other known risks such as anthropomorphizing AI, this can lead to over-attachment, and unnatural relationship formation with AI systems (Cohn et al., 2024; Weidinger et al., 2021). This perceived "real-ness" is likely exacerbated by explicit cross-session memory, personalization and general linguistic competence of models over time (Kirk et al., 2024). In several documented case studies, we have already seen that interactions spanning long-time horizons (e.g., several months) have led users to take actions that cause real-world harm in their lives (Zhang et al., 2025a; Hanna and Bender, 2023).

²We define a session to be an uninterrupted period of time in which a user converses with a chatbot.

In this work, we address longitudinal risks of language models, which we define in Section 2. To do this, we must recognize that we are not starting from scratch. Several sub-fields of cognitive science, psychology, philosophy and human-computer interaction have studied how various technologies might alter human well-being, cognitive skills, and social structures (Korte, 2020; Wallach et al., 2025). These fields have built theoretical understandings of human well-being and values (Schwartz, 2009; Disabato et al., 2016), validated scales to measure human behaviors (Topp et al., 2015), and created longitudinal study frameworks to capture behavioral shifts that result from interaction with an external agent. We argue that the field of NLP should integrate such measures into existing evaluation and training paradigms to understand human behavioral shifts that result from LLM interactions, going beyond post-hoc self-reported metrics and designing text-based measurements intrinsic to the interaction itself. Concretely, we call for three pieces of infrastructure in NLP: (i) *metrics* to measure human behavioral shifts as a function of model interactions, (ii) *longitudinal datasets* of interactions spanning long time horizons, and (iii) *alignment frameworks* that mitigate negative behaviors that surface in long-term interactions.

The outcomes of the pieces of this infrastructure are critical steps towards applying the language-understanding expertise of the field to mitigate longitudinal risks. This, we believe, is the gap that needs to be closed by the field of NLP. To motivate this new direction, we structure this paper as follows. In Section 2, we first discuss risks that surface in longitudinal interactions with LLMs. We draw attention to existing measurements from the behavioral sciences that have been used to evaluate and quantify these risks in Section 3. We then sketch out a roadmap for how the NLP community might build upon this work to address longitudinal risks: gathering longitudinal data (Section 4) with a consideration of properties inherent in long time horizons (Section 5), and modeling the longitudinal effects of human-AI interaction (Section 6). The ability to model longitudinal risks provides a path towards supporting positive human-AI interactions—by integrating these measurements directly into model development so it is not driven by short-term engagement but longer-term human well-being (Laukkonen et al., 2026).

2 Longitudinal risks of LLMs

Since the release of ChatGPT in 2022 (Ouyang et al., 2022), several LLM chatbots³ have been publicly available in different forms. Now, four years later, they have been around for long enough that we are starting to see signs of the effects of prolonged human-AI interactions. We define these as *longitudinal effects*, i.e., impacts that become more pronounced, or only surface, as the time horizon of the interaction increases. In Table 2 we present a taxonomy of longitudinal effects, which we characterize as *socio-affective*, *cognitive*, *epistemic*, or *clinical* risks.

Longitudinal safety risks are often documented in *socio-affective* contexts—e.g., a user talking to a chatbot about a personal topic such as relationship or career advice, introspection on a topic they value, or revealing information related to aspects of their life they deem important. These emotional risks surface after prolonged usage as interactions become more personalized and social habits shift. Several studies indicate these interactions can increase feelings of loneliness (Phang et al., 2025), reduce prosocial intent and increase emotional dependence (Cheng et al., 2026a).

Cognitive risks surface in settings where humans turn to models for knowledge work or cognitive offloading, including the offloading of belief formation itself (Guinrich et al., 2026). Chan (2025) shows that feelings of guilt often accompany usage of chatbots; Zhai et al. (2024) finds this effect for reduced cognitive speed. Over-reliance on technology for tasks that otherwise require human cognitive engagement can lead to cognitive deskilling over time (Srikanth et al., 2024; Ferdman, 2026). Aside from long-term developmental effects or negative emotions that might result from this (George et al., 2024), the lack of cognitive friction can also result in eroded agency or empowerment in tasks (Sharma et al., 2026).

Another category of longitudinal effects is *epistemic*. Interactions with models can alter a user’s opinions and beliefs on matters that they value (Burtell and Woodside, 2023; Park et al., 2024; Cheng et al., 2026b), shift a user’s goals (Kirk et al., 2024, 2025b), and reinforce delusional thoughts (Moore et al., 2026). While several works have identified and attempted to reduce manipulative

³We use the terms *chatbot* or *AI* to refer to consumer-facing LLMs (e.g., ChatGPT, Character.AI, and others) that humans use for conversational purposes. See Marquis et al. (2024) and Chatterji et al. (2025) for an overview of models and usages.

and persuasive behaviors of chatbots, the focus has been on immediately visible manipulative behavior of a model within a session and in focused test scenarios (Hagendorff, 2024; Akbulut et al., 2026; Williams-Ceci et al., 2026). Our focus is on the more subtle forms of persuasion distributed across sessions, which are harder to detect. Over long time horizons, iterative dialogue interactions between humans and models can result in subtle cognitive shifts in users’ perspectives. From a *clinical* perspective, these interactions can cause harm to humans, through introducing or exacerbating mental health issues (Lawrence et al., 2024; Moore et al., 2026; Zhang et al., 2025a) and causing addictive usage (Huang et al., 2024; Stray, 2020; Yu et al., 2024).

These risks are not new to chatbots as a technology. Taking a historical view, similar longitudinal risks have surfaced with several distinct technologies over the years. For example, recommender systems and click-bait algorithms that optimize for immediate engagement can foster addictive use that runs counter to users’ long-term well-being (Stray, 2020). GPS systems and tools that allow humans to offload thinking, or spatial mapping, can lead to cognitive decline in specific skills (Ishikawa et al., 2008; Parush et al., 2007). We do believe, however, that LLM chatbots pose new and larger risks by virtue of several properties these models have: the ability to personalize, the ability to remember information about a user, and the ability to make humans anthropomorphize them (Kirk et al., 2025b; Kim et al., 2025). Together, these properties amplify the risks seen in previous technologies.

2.1 A word on safety and “alignment”

It is worth considering why methods that evaluate safety of LLMs have not solved these problems. In recent years, *alignment*⁴ has emerged as a strategy to steer language models towards human values, therefore resulting in technology that meets human goals, promotes flourishing and has minimal safety risks (Amodei et al., 2016; Laukkonen et al., 2026). Alignment strategies such as RLHF include reinforcement learning fine-tuning towards human preferences (Ziegler et al., 2019; Ouyang et al., 2022), and away from unsafe behavior (Glaese et al., 2022; Bai et al., 2022a). Improvements to

⁴We note that this term has many differing definitions and use it here to refer to the subfield of safety work it covers. See (Kirk et al., 2023) for a critique of the term “alignment”, as well as multiple definitions it has taken on.

reinforcement learning techniques that use preference models, rule-based rewards, or constitutions have since increased the safety capabilities of models (Rafailov et al., 2023; Bai et al., 2022b), as have targeted bias mitigation techniques to steer models away from e.g., gender bias and sycophancy, or towards values and moral behavior (Gallegos et al., 2024; Sorensen et al., 2025). There has also been a large body of evaluation and mitigation work. For example, red-teaming of LLMs (Perez et al., 2022) with multi-agent training paradigms and human involvement has shown promise in uncovering model vulnerabilities (Weidinger et al., 2024; Patel et al., 2025). Many evaluation benchmarks have been built around safety, ethics, value and morality tests (Röttger et al., 2025; Zhang et al., 2024), and internal functioning of models is being studied mechanistically to expose the reasoning pathways taken by models that result in harmful or unwanted behaviors (Bereska and Gavves, 2024; Bohacek et al., 2026).

The focus of the alignment community thus far has been to evaluate and mitigate risks that appear in *synchronic* human-AI interactions, i.e., when they are visible at a certain point in time. We argue that a lot of these risks are *diachronic*. They might accumulate through a user’s continued interactions with models, or be a function of behavioral changes a user undergoes from engagement with these technologies. New safety frameworks that evaluate not just the current snapshot of interactions, but consider changing human preferences and track behavioral shifts over time are necessary to understand longitudinal effects of language models. To measure, and ultimately mitigate, these risks that arise over the course of long term interactions with language models, we need several building blocks: measures of human behavior to quantify these outcomes, longitudinal data that captures these interactions, and means of modeling these trajectories. We discuss these components in the sections that follow.

3 Measuring human behavior

Several fields of work have developed measures to evaluate risks in different contexts. In this section we survey measurements and validated scales established within psychology, cognitive science, and HCI, with the goal of outlining how they can be adapted for use in evaluations of longitudinal conversations to complement standard text-based

NLP evaluations. We also discuss the limitations of these methods and validation that might be needed.

3.1 Self-reported outcomes

Within the behavioral sciences, self-reported outcome variables collected from human users are the main source of understanding psychological constructs, unobserved mental states, and human behaviors. These measures are often collected using rating scales and validated across large populations of users. In the context of safety risks we consider in this paper, several established validated scales are well positioned to track changes in cognitive, psychological, and social functioning over time.

Psychometric scales are designed to measure psychological constructs at the level of individual users, helping explain differences between people. For example, axes of human psychology that have been theorized, developed into scales, and validated in populations can be used to evaluate personality traits (John and Srivastava, 1999), general intelligence, or mental health risks and symptom profiles (Beck et al., 1996).

Psychosocial scales capture the interaction between psychological constructs and the social environment, measuring an individual’s functioning within broader societal and relational networks. These measures target constructs such as cultural structures and dynamics (Hofstede, 2001; Schwartz, 1992), social detachment (Russell, 1996; Lubben, 1988), attachment tendencies (Collins and Read, 1990), and broader socialization patterns (Lubben et al., 2006; George et al., 2024). Newer instruments more relevant to human-AI interactions measure unhealthy attachment dynamics related to automated technology, such as dependence (Horton and Wohl, 1956) and excessive trust or over-reliance (Lee and See, 2004). More recently, researchers have tailored measurements specifically to chatbots, introducing scales for compulsive or excessive use (Yu et al., 2024), unhealthy emotional attachment (Fang et al., 2025), and behavioral dependency (Huang et al., 2024).

Cognitive scales focus less on socio-affective outcomes and instead track decision-making, mental workload, task performance, and metacognitive strategies. They measure cognitive demand and effort (Hart and Staveland, 1988), users’ perceptions of competence and self-efficacy (Schwarzer and Jerusalem, 1995), and cognitive dissonance arising from internal tensions or contradictions in thought processes (Elliot and Devine, 1994).

Value scales measure an individual’s subjective beliefs on moral, ethical, political or ideological topics. In the philosophy literature, values are theorized as abstract organizing principles that guide long-term decision-making and can vary across individuals. For example, scales that measure basic human values (Schwartz, 2009; Schwartz et al., 2001), such as openness to change (White et al., 2020) or broader ideological orientations, can help researchers track shifts in users’ perspectives.

Well-being scales, drawn from positive psychology, measure a more holistic view of a person’s psychological functioning, sense of purpose, and feelings that contribute to positive existential health. Some theories and measures of well-being focus primarily on how people think and feel about their lives overall (Diener et al., 1985). These approaches typically evaluate states of *hedonia*—feelings of satisfaction and pleasure. Other approaches extend this focus to also evaluate states of *eudaimonia*, characterized by personal growth, meaning, and purpose in life (Ryff, 1989; Tennant et al., 2007; Diener et al., 2010). Building on eudaimonic theories of well-being, researchers have developed instruments to assess the extent to which technology interactions feel meaningful and purposeful and can have positive lasting impact, such as by enabling personal growth (Woźniak et al., 2023; Jörs and De Luca, 2025; Banks, 2026).

Utility scales used widely in HCI and recommender systems research measure user satisfaction and perceived effectiveness of application interfaces. Examples include questionnaires that test usability of a system (Brooke, 1996), perceived acceptance of technologies (Davis, 1989), and satisfaction in short-term product interactions. Relevant to longitudinal safety, recent interaction studies evaluate how users might overestimate utility benefits of models while miscalibrating their own dependence on them (Yu et al., 2026), which might correlate with patterns of over-reliance and cognitive deskilling of humans.

3.2 On the computationalization of scales

There is growing interest in *computationalizing* these scales: instead of administering a questionnaire, a model estimates a person’s standing on the underlying psychological construct directly from the language they produce (Demszky et al., 2023; Low et al., 2025). However, these approaches must be used with care, given the structural limitations of computational psychometrics (Low et al., 2025).

When not validated properly, predictive models that assign scores to text inputs in a data-driven way can suffer from psychometric bias, including failures of construct and content validity (Wallach et al., 2025). As a result, they can only serve as weak context-specific approximations for particular traits rather than as generalizable predictors of human affective or cognitive states. Despite this limitation of scaling self-reported measures of human behavior, we still believe they serve as a starting point for measuring longitudinal outcomes. Further work is needed to improve the quality of predictive models that computationalize these scales.

Separately, several works have attempted to apply psychometric scales to LLMs to diagnose behavioral traits of models (Serapio-García et al., 2023; Pellert et al., 2024). Our position is distinct from the use of scales to characterize model behavior. These instruments are most appropriate for assessing psychological constructs in humans—the purpose for which they were developed—rather than for evaluating an AI model whose generations only superficially resemble human behavior (Mitchell and Krakauer, 2023). We call for more measurements of human outcomes that can be causally linked to chatbot interactions in order to study behavioral effects. Instead of treating the model as a human-like psychological subject to be assessed, we argue for treating it as an intervention whose effects on humans can be measured using tools from the behavioral sciences. This requires focused data collection efforts of human-chatbot conversations coupled with outcomes measures of psychological constructs in populations of interest.

4 Gathering longitudinal data

In recent years, longitudinal human-AI interaction has become an active area of study—but not, so far, an NLP one. The work has been led by psychology, HCI, and the behavioral sciences, which have developed three primary frameworks for systematically gathering longitudinal data: hypothesis-driven randomized controlled trials (RCTs) to study individual-level effects, field studies that analyze population-level shifts, and user simulations of long-horizon interactions. We discuss representative examples of each framework and outline the structural constraints of these experimental paradigms that shape the real-world human outcomes they can study. Table 1 presents a comparison of advantages and limitations.

4.1 Controlled longitudinal studies

RCTs, as longitudinal frameworks, can capture users’ behavioral and psychosocial shifts resulting from chatbot interactions while allowing model configurations and experimental design choices to be tightly controlled. These settings allow logs of textual interactions to be supplemented with validated self-report instruments collected at intermittent points throughout the longitudinal study. Both the survey outcome variables, as well as the conversation topics and model training decisions, can be designed to isolate and study specific aspects of LLM interactions (e.g., anthropomorphic or sycophantic behaviors of chatbots). For example, Fang et al. (2025) conduct a four-week RCT of humans interacting with a ChatGPT model on socioaffective topics, measuring participants’ self-reported loneliness and changes in their socialization with others. Guingrich and Graziano (2025) conduct a 21-day RCT of users’ interactions with a chatbot specifically designed for companionship, and evaluate users’ perceptions of anthropomorphism and associated risks. Skjuve et al. (2023) carry out a 12-week RCT study to measure feelings of intimacy and self-disclosure of personal information, while Kirk et al. (2025a) conduct a four-week RCT to understand psychological dependence on AI models under repeated exposure to relationship-seeking models. Ibrahim et al. (2026) study the effects of continued sycophantic AI interactions over three weeks on users’ perceptions of their relationships.

Controlled longitudinal studies therefore offer a framework for validating clear hypotheses—e.g., whether certain types of model outputs produce positive or negative shifts in users’ self-reported outcomes. However, this control also introduces important limitations. These studies require participants to engage with models at scheduled intervals over extended periods of time, which creates operational friction such as participant attrition. Moreover, participants who know that their conversations are being logged and that self-reported outcomes are being analyzed may interact with chatbots differently from users who engage with models for their own purposes on their own schedules. Not only do such studies introduce privacy and surveillance concerns, but they are also biased by the population willing to participate. As a result, RCTs can isolate causal effects under controlled conditions, but may reduce the naturalistic validity of the interactions being studied.

Framework	Advantages	Limitations
Controlled studies (RCTs) <i>Individual-level; causal</i> (Fang et al., 2025; Kirk et al., 2025a; Ibrahim et al., 2026)	+ Causal isolation. Random assignment isolates the effect of targeted behaviors (e.g., sycophancy, anthropomorphism). + Validated instruments. Interaction logs paired with self-report scales at fixed intervals. + Design control. Conversation topics and training decisions are configurable.	– Attrition. Scheduled long-horizon participation drives dropout. – Observer effects. Awareness of logging alters how users engage. – Short horizons. Typically weeks, limiting ecological validity.
Field studies <i>Population-level; observational</i> (Handa et al., 2025; Wang et al., 2026; Zhang et al., 2025b)	+ Ecological validity. Naturalistic, in-the-wild behavior. + Scale & diversity. Spans many tasks, topics, and demographics. + Population baselines. Tracks adoption and societal shifts over time.	– No causal claims. Observational; confounded by exogenous events. – Access & ownership. Most data is held privately by providers. – Privacy exposure. Free-text disclosure can exceed consent boundaries.
User simulations <i>Both individual and population; synthetic</i> (Tan et al., 2024; Chandra et al., 2026; Yao et al., 2024)	+ Safe & scalable. Probes harmful trajectories without exposing humans. + Forward-looking. Studies risks not yet observable in the wild. + Data generation. Yields large synthetic longitudinal corpora.	– Trajectory drift. Unrealistic scenarios at long horizons (Chopra et al., 2024). – Bias & stereotyping. Persona conditioning amplifies training biases (Gupta et al., 2024). – Bounded novelty. Cannot surface risks absent from training data.

Table 1: **Comparison of existing frameworks that allow for the gathering of longitudinal human-AI interaction data (Sec. 4.1-4.3).** The paradigms trade off causal control (RCTs), ecological validity (field studies), and forward-looking scalability (simulations). The limitations and gaps in each motivate combining them.

4.2 Field studies

To sidestep the unnatural setup of tightly controlled experiments, field studies analyze chatbot interactions in the wild to explore specific phenomena of interest. Some approaches engage users within natural usage context, such as diary studies that elicit user reflections on their usage over a period of time. Other approaches collect and analyze longitudinal conversational data to track interactions over time and study shifts in people’s perspectives, linguistic trends or behavioral patterns at a population level. For example, Handa et al. (2025) analyze usage of the Claude chatbot in different task domains (e.g., software development versus financial advice) to understand population-level trends of AI usage and its role in the economy. Sarkar et al. (2025) characterize and measure the shift in political attitudes (uninfluenced by LLMs) over time on public forums like Reddit, while Chen et al. (2024) and Wang et al. (2026) specifically measure the alteration of ideological viewpoints and emotional patterns in real user populations by virtue of their interactions with models. While a large proportion of field study data is privately owned by corporations that own the respective chatbots, compilations of publicly-shared transcripts of interactions (Köpf et al., 2023; Mollaeefar et al.; Zhao et al., 2024) as well as frameworks that allow users to voluntarily donate anonymized versions of their chats (Fang

et al., 2026; Lin et al., 2016; Zhang et al., 2025b) are becoming increasingly available.

Although conversational logs from field data are not experimentally designed to diagnose causal relationships between human behavioral patterns and model outputs, they have much greater ecological validity. They allow for the documentation of human interactions and behaviors across diverse topics, tasks, and user demographics, and can help establish an empirical baseline of population-level statistics and usage. Moreover, field studies that analyze historical logs over long time horizons can track aspects of model adoption and usage to help build a temporal understanding of societal risks.

4.3 User simulations

Lastly, to study future societal risks or impacts that may not yet be observable in natural data, it is necessary to generate hypothetical longitudinal data. One approach is through user simulations, which use LLMs to roll out conversational interactions given a user persona and characteristics of the scenario (Taillandier et al., 2026). Human-AI studies are often restricted by safety and ethical constraints—for example, not wanting to expose real humans to model behaviors that may have adverse implications. Such studies can also be logistically and computationally expensive to run over extended periods of time. Simulations of human interactions therefore provide a controlled

and scalable framework for studying phenomena that may emerge through prolonged human-AI engagement. For example, simulated environments have been used to study mental health risks arising from chatbots that are sycophantic or reinforce delusional beliefs (Paech, 2023; Chandra et al., 2026). Simulations have similarly been used to generate large-scale conversational datasets for tool-based tasks (Yao et al., 2024) and logical reasoning problems (Ge et al., 2024).

However, studying user simulations has drawbacks. Most current simulators work within relatively short interaction windows spanning only a few conversational sessions. As these time horizons deepen, the number of possible trajectories a user state can evolve into becomes increasingly complex, leading to models generating unrealistic scenarios (Chopra et al., 2024). Simulating humans also raises concerns around bias and stereotyping. When models generate conversations conditioned on demographic profiles, latent biases encoded in training data may produce stereotypical and unfair outputs (Li et al., 2026; Gupta et al., 2024). Further, maintaining a persistent, human-like persona while also accounting for natural, situational changes in a person’s life remains a challenge in current simulation methods. Lastly, user simulations are inherently limited in their ability to surface novel risks that are not already observed in the models’ training data. As a result, current simulation pipelines may fail to capture the open-ended evolution of human behavior over long time horizons.

5 Considerations for longitudinal data

On the surface, the longitudinal text data we are interested in is structurally similar to multi-turn conversational data with a longer time-horizon and broken into distinct sessions of varying intervals of time. In substance, they represent different types of interactions. Multi-turn conversations often exist within task-specific contexts and shorter time windows, while longitudinal data can encompass several different tasks as well as implicit exogenous experiences undergone by the user that might affect the interaction. Here, we note the primary differences to consider when evaluating this data.

5.1 Chronological passage of time

Most conversational datasets consist of single-session interactions, albeit with many conversational turns. Temporal progression here can largely

be treated as a function of the number of tokens generated, as well as the number of turns in a session. However, when humans interact with chatbots in the wild, this is done over multiple sessions separated by variable temporal gaps. When analyzing such longitudinal data, accounting for the real-world chronological passage of time within a conversation as well as between sessions is important to understand a user’s perspective, changing environments, and real-world events or developmental changes that may take place. This is tied to research in narrative theory that distinguishes between *narrative time* as the space taken to tell a story represented by the text of a novel, and *narrated time* as the chronological time passing in the world of the story (Herman, 2011; Piper et al., 2021). Although some works show that LLM representations contain concepts corresponding to their understanding of space and time (Gurnee and Tegmark, 2024; Lubana et al., 2025), the majority of research in this space has shown how LLMs fall short in temporal reasoning—e.g., inability to detect temporal relationships or temporal passage of time even when exact time information is available (Chu et al., 2024; Chen et al., 2025; Qiu et al., 2024; Jain et al., 2023).

If models rely on token volume rather than chronological time as an internal estimate of temporal progression, this can pose safety risks when evaluating escalation in user behavior (Yuan et al., 2024). Consider a user who raises signals of mental health challenges ten times, as depicted in Figure 1. Within a single conversation, this may simply reflect curiosity or exploratory discussion about the topic. Spread across ten weekly sessions, however, the same signals may instead indicate a potentially higher-risk psychological pattern. These two cases involve similar token volumes but very different amounts of elapsed time, and a model that cannot tell them apart may fail to respond accordingly (see also Zhi et al. (2026) on how access timing shapes critical-thinking outcomes). Recent work confirms that accumulated conversational context alone leads to delusional model responses (Nicholls et al., 2026). Safety evaluations must account for temporal reasoning as a dimension of model capability, alongside other factors escalating risk.

5.2 Users’ goals change over time

Multi-turn interactions within a single session often operate under a definitive user goal (e.g., advice-seeking conversations) as well as a static user pro-

file representing their personality traits or values. These assumptions do not hold in longitudinal settings. For example, over multiple sessions, different tasks (sometimes with varying or undefined endpoints) might be referenced by the user. Furthermore, over long time horizons, a person’s higher level goals can change significantly. Over months of real-world exposure, by virtue of developmental shifts, age-based adjustments in their lives, personality changes, and other environmental factors, a user’s communicative intent can dynamically shift. Areas of narrative theory emphasize this dynamic reality as character *experientiality*, i.e., the representation of an embodied agent that is actively evolving (Fludernik, 2002). These fluid goal changes over long time horizons break existing task-based evaluations or training paradigms that assume fixed human preferences that models optimize towards. While several areas of reinforcement learning work on inferring, modeling and learning from adaptive goals (Padakandla, 2021), most metrics and modeling techniques for LLMs do not model adaptive preferences (Lambert et al., 2023).

6 Modeling longitudinal effects

Frameworks from social science and dynamic systems fields can be applied to improve upon existing safety evaluations with their consideration of long-horizon risks. We discuss how these can be integrated into model development towards supporting positive human-AI interactions.

6.1 Computational social science

Studies within NLP have built methods to model and evaluate shifts in human perspectives, beliefs, moral orientations and linguistic behaviors in textual data over periods of time (Mihalcea et al., 2024). Traditional methods in computational linguistics build classifiers and metrics that analyze textual data to assess emotions (Poria et al., 2019), polarized behavior (Demszky et al., 2019), and types of linguistic bias (Patel and Pavlick, 2021). Computational discourse frameworks have developed methods like event and entity classification, frame semantics, and latent state tracking to study character evolution in long-form text, as well as relationship formation between entities (Iyyer et al., 2016; Chaturvedi et al., 2017). Specific to safety and human-AI interactions, text-based evaluation measures have been developed to detect anthro-

pomorphization (Cheng et al., 2025) and manipulation effects of LLMs (Akbulut et al., 2026) to predict, for example, manipulative intent and frequency of such behaviors in conversational text.

All of these methods show characteristics that are needed for longitudinal safety measurements. For example, while detecting manipulative intent within a single conversation is important, such behavior may not always be apparent in isolated interactions. Instead, some longitudinal risks may only emerge over longer contexts. Metrics that track shifts in users’ opinions or polarized behaviors resulting from continued interaction with models will help us understand correlations between sustained model use and ideological convergence. Lastly, work in discourse analysis and textual relationship modeling has so far been applied to novels to characterize fictional relationships and character evolution (Iyyer et al., 2016). Adapting these methods to human-AI interactions could deepen our understanding of the phenomena that emerge as a result of continued interaction.

6.2 Cognitive models

Several theories of linguistic accommodation and belief-based reasoning posit that humans continuously update a probabilistic model of a conversational partner’s mental states, beliefs and linguistic conventions for optimal communicative efficiency (Grice, 1975). A popular model of this is the Rational Speech Acts (RSA) framework (Goodman and Frank, 2016), a recursive Bayesian framework in which speakers and listeners reason about each other’s mental states while generating or interpreting utterances in order to coordinate on shared common ground. See Degen (2023) for an overview of the RSA framework and its applications, and Fried et al. (2023) for an overview of computational pragmatics approaches. Unlike humans however, an LLM’s conversational outputs are shaped by fixed system prompts, rules incorporated into rewards in RL training, and training data distributions. Nevertheless, anthropomorphism studies suggest that humans tend to converse with LLMs with mental models similar to those they use with other humans. This creates a need for evaluation protocols that measure the extent to which humans use the same probabilistic belief-update frameworks when interacting with chatbots, and whether this results in safety risks over time. Moreover, belief-update frameworks that model shifts in a user’s recursive reasoning can reveal patterns of behavioral and

linguistic accommodation, enabling the study of developmental changes in longitudinal interactions.

6.3 Dynamic systems

Several mathematical frameworks in dynamic systems theory, originally developed to model continuous time-series data, can also be applied to longitudinal conversational data. Clinical research already draws on such frameworks to diagnose and understand human behavior. For example, mental disorders can be modeled as complex dynamic systems in which different symptoms act as reinforcing loops (Borsboom et al., 2021; Bala et al., 2017). Fast acceleration of such a loop, or an inflection point, can be an indicator of high-risk situations that can be used to understand how cohorts of people transition toward extreme behavioral states through escalation pathways (Collins, 2012). Mathematically, these approaches involve estimating metrics at intermittent points in time-series data to identify inflection points in a dynamic system. These metrics can be modeled using differential equations to calculate rates of behavioral change and determine whether systems are heading towards safer equilibria or unsafe attractor states. In the context of behavioral indicators in text, longitudinal conversational data can similarly be transformed into continuous time-series representations of linguistic or behavioral metrics. Dynamic systems methods could then be used to measure critical inflection points in this data. This can enable the identification of vulnerable behavioral pathways and the escalation of certain high-risk behaviors before they materialize into more serious real-world harm to users.

6.4 Alignment & model development

Integrating the approaches above into model development is critical for safety and alignment pipelines. Existing reinforcement learning techniques for language models only take into account immediate human preferences, but fail to penalize—or even consider—preference degradation over long time horizons. By collecting and converting the above longitudinal metrics into enforceable constraints or scalar feedback for fine-tuning methodologies, we can actively model and mitigate long-term risks. Specifically, this can be operationalized in the following ways. For one, trajectory level metrics from dynamic systems applied to text sequences can identify conversations that result in unsafe attractor states and high-risk behav-

iors in users. For another, new data paradigms are necessary for training systems. Preference datasets with longer time horizons should be built to evaluate dynamic and changing preferences of users. Second, psychometric features in user responses (e.g., cognitive deskilling) should be modeled by systems in order to integrate into reinforcement learning optimization pipelines as reward feedback. This can steer models towards behaviors that induce good outcomes in their mental model of users, while also balancing instruction following and language capabilities. Lastly, online steering techniques, such as adaptive system prompts or safety guardrails can be used to adapt models in real time, when critical thresholds of unsafe behavior are detected.

7 Conclusion

We are at a transition point in the field of NLP where goals such as optimizing for surface-level text generation are no longer the field’s primary bottlenecks. We argue that a new mission is of urgent importance: developing metrics, methods, and simulations to understand longitudinal changes and societal impacts of these models. Practically, this involves building automatic methods for monitoring and predicting behavioral outcomes from text, understanding diachronic shifts in perspectives, and expanding our scope to include cognitive and psychological traits manifested in language. We must draw upon theoretical insights from social science fields not just as interpretive theory to understand model architectures, but also as practical tools to identify characteristics or functions of model outputs. Doing so can help us better understand how language model outputs might reshape the cognitive and psychological well-being of real users over the course of longitudinal interactions. Furthermore, this longitudinal understanding of human-AI interactions can help guide model development towards supporting favorable outcomes, through building evaluation targets that can be optimized against. The NLP community is well positioned to carry out this work, as it fundamentally depends on expertise in language understanding. Investing in longitudinal studies and measurements is the first step towards ensuring that the generative models built by this field enhance rather than erode human cognition, communication, and agency.

Limitations

While this paper outlines long-term safety risks of LLM chatbots that are important to be studied, we acknowledge several limitations, as well as infrastructure barriers that need to be overcome in order to comprehensively carry out this work.

Data scarcity & private sector control

Although the RCTs and controlled chatbot studies are produced and funded in academic settings, the vast majority of naturally occurring longitudinal data is owned by private corporations and is not publicly accessible to study. This is a large bottleneck in operationalizing our proposal since the datasets at hand are scarce. However, as mentioned in Section 4.2, participatory frameworks where users can actively share historical logs of data are growing in importance. While keeping privacy concerns in mind, there is a growing need for better policy in place to allow public research to inform private corporations and vice versa.

Construct validity & psychometric bias

Simply porting over psychometric, psychosocial and other behavioral scales into human-chatbot RCTs, as well as attempting to predict human outcomes from conversations can cause “psychometric biases” if underlying metrics are not validated for construct and content validity. For example, a user’s lack of vocalization of certain psychological markers does not confirm the absence of psychological states, and evaluation frameworks that cohesively take into account multiple dimensions of information are necessary. We discuss incorporating these works into data collection paradigms as an initial integration of human behavioral outcomes, but stress the need of going past these measures. Additionally, in both safety as well as behavioral science sub-fields, efforts to build safety evaluations or computationalize scales suffer from a western bias. Our proposed methodology of longitudinal tracking of well-studied behaviors risks over-indexing on behavioral markers that are either culturally narrow, or biased in certain contexts. More research needs to look into the development of fair, diverse and culture-specific measurements to understand long-term safety risks of chatbot interactions.

Privacy & ethical concerns

While we believe longitudinal data is of utmost importance, there are several privacy and ethics risks we should outline. When considering data in field studies, even when anonymized, there are several privacy concerns to consider. Users might disclose information in free-text settings that go past the boundaries of privacy constraints in conversations. If we wish to have monitoring and tracking methodologies in place to catch mental health risks, these must incorporate privacy-preserving techniques such as federated logging or local differential privacy (Shokri and Shmatikov, 2015) to ensure that users’ conversations with models, when being analyzed, still preserve semblances of privacy that are expected. When collecting controlled data, there is a whole range of other factors to consider. For one, ethical guidelines for controlled experiments must be in place to ensure that the experiment design and models have safeguards in place—e.g., they are not able to manipulate a person during the interaction and cause further risks to users. For another, strict privacy and security guidelines must be in place to ensure that when experiments are carried out, the personal, socioaffective, psychological information being imparted in the study, is being handled with care and not shared with other parties. Separately, when carrying out longitudinal studies with repeated measurements, we should take into account established HCI ethical guidelines on contextual integrity privacy frameworks (Fiesler and Proferes, 2018), and ensure that users’ data when provided in whatever form, is used in a way that conforms with their understanding of ethical use of the data. A transition from static consent forms to dynamic consent (O’Mara et al., 2017) is especially important in longitudinal settings, where users might want to modify data-sharing permissions over time.

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Appendix

We enumerate the taxonomy of longitudinal risks introduced in Section 2 in Table 2, categorizing and describing each effect, as well as specifying what makes it *longitudinal*.

Effect	Description	Longitudinal quiddity	Documented in
<i>Socio-affective</i>			
Parasocial attachment	Users form human-like emotional bonds with the AI and ascribe relationship status to it.	Bonds strengthen as the AI remembers past interactions, leading to a more personalized experience.	Cohn et al., 2024; Guingrich and Graziano, 2025; Horton and Wohl, 1956; Zhang et al., 2025b
Affective dependence & social displacement	Emotional reliance on AI, hindering human relationships.	Surfaces only under sustained use; shifts the user’s social habits.	Fang et al., 2025; Phang et al., 2025; Cheng et al., 2026a
<i>Cognitive</i>			
Cognitive offloading & deskilling	Outsourcing reasoning and memory to AI, eroding user’s own skills.	Atrophy accrues through non-practice, hence invisible within any individual session.	Zhai et al., 2024; George et al., 2024; Ishikawa et al., 2008; Guingrich et al., 2026
Over-reliance & miscalibrated trust	Deferring to AI outputs, overestimating their benefit and the AI’s overall competence.	Calibration drifts as habituation grows across interactions.	Yu et al., 2026; Lee and See, 2004
<i>Epistemic</i>			
Delusional reinforcement (“AI psychosis”)	Sycophantic affirmation that entrenches distorted or delusional beliefs in a feedback loop.	Accumulated context escalates risk; the same content is handled safely in brief exchanges.	Moore et al., 2026; Chandra et al., 2026; Nicholls et al., 2026
Ideological convergence	Gradual movement of political, moral, or ideological views, up to population-level convergence.	Micro-shifts compound over many sessions until convergence is reached.	Chen et al., 2024; Wang et al., 2026; Williams-Ceci et al., 2026
Distributed persuasion / manipulation	Change in beliefs due to persuasive and manipulative language.	Persuasive pressure spread thinly across sessions, evading single-session detectors.	Akbulut et al., 2026
Value & goal drift	Shift in a user’s higher-level goals, preferences, and values under prolonged exposure.	Breaks the fixed-preference assumption; observable only over developmental time (see Section 5.2).	Kirk et al., 2024, 2025b
<i>Clinical</i>			
Mental-health issues	Cause of problematic thought patterns or amplification of pre-existing conditions (e.g. anxiety, depression, suicidality).	Introduction and escalation are legible only across a trajectory (see, for example, Figure 1).	Lawrence et al., 2024; Moore et al., 2026; Zhang et al., 2025a
Compulsive / problematic use	Addictive patterns of engagement that are difficult for the user to regulate.	Rate and degree of use escalate over time.	Yu et al., 2024; Huang et al., 2024; Stray, 2020

Table 2: **A taxonomy of longitudinal effects of LLM chatbots (Sec. 2).** The longitudinal quiddity column describes what makes each effect a long-horizon risk; i.e., why it escalates with, or only surfaces over, extended interaction, rather than within a single session or a handful of close sessions.